INSTRUMENT CLUSTER (Continued)

this layer is also translucent. The darkness of this outer layer prevents the cluster from appearing cluttered or busy by concealing the cluster indicators that are not illuminated, while the translucence of this layer allows those indicators and icons that are illuminated to be readily visible. The underlying layer of the overlay is opaque and allows light from the LED or incandescent lamp for each of the various indicators and the incandescent illumination lamps behind it to be visible through the outer layer of the overlay only through predetermined stencil-like cutouts. A rectangular opening near the base of the overlay between the speedometer and tachometer dial faces has a smoked clear lens through which the illuminated VFD unit can be viewed.

Several versions of the EMIC module are offered on this model. These versions accommodate all of the variations of optional equipment and regulatory requirements for the various markets in which the vehicle will be offered. The microprocessor-based EMIC utilizes integrated circuitry and information carried on the Programmable Communications Interface (PCI) data bus network along with several hard wired analog and multiplexed inputs to monitor sensors and switches throughout the vehicle. In response to those inputs, the internal circuitry and programming of the EMIC allow it to control and integrate many electronic functions and features of the vehicle through both hard wired outputs and the transmission of electronic message outputs to other electronic modules in the vehicle over the PCI data bus. (Refer to 8 - ELECTRICAL/ELECTRONIC CONTROL MODULES/COMMUNICATION - DESCRIPTION - PCI BUS).

Besides typical instrument cluster gauge and indicator support, the electronic functions and features that the EMIC supports or controls include the following:

- **Audible Warnings** - The EMIC electronic circuit board is equipped with an audible tone generator and programming that allows it to provide various audible chime tone alerts to the vehicle operator. (Refer to 8 - ELECTRICAL/CHIME/BUZZER - DESCRIPTION).
- **Audio System Cabin Equalization** - The EMIC stores the cabin equalization curves for various optional speaker system architectures for use by the radio. The EMIC provides this information when requested by the radio over the PCI data bus. Changing the cabin equalization settings of the EMIC requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.
- **Axle Locker Control** - The EMIC provides the logic that controls the operation of the rear and front axle locker feature found on models equipped with the optional off-road package.
- **Electronic Pinion Factor** - The EMIC stores the tire size, axle ratio, and transfer case type information for use in calibrating the proper vehicle speed and distance values for display by the speedometer and odometer, as well as for use by other electronic modules in the vehicle. The EMIC provides this information when requested by other modules over the PCI data bus. Changing the electronic pinion factor settings of the EMIC requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information.
- **Interior Lamp Load Shedding** - The EMIC provides a battery saver feature which will automatically turn off all interior lamps that remain on after a timed interval of about twenty minutes.
- **Interior Lighting Control** - The EMIC monitors inputs from the interior lighting switch and the door ajar switches to provide courtesy lamp control. This includes support for a timed illuminated entry with a theater-style fade-to-off feature.
- **Panel Lamps Dimming Control** - The EMIC provides a hard wired 12-volt Pulse-Width Modulated (PWM) output that synchronizes the dimming level of all panel lamps dimmer controlled lamps with that of the cluster illumination lamps. This includes providing features such as VFD unit illumination when a door is ajar (rental car mode), radio illumination control, a parade mode, and one step dimmable front and rear lock, cruise, and four-wheel drive indicators.
- **Rear Window Defogger Control** - The EMIC provides control and timer functions for the output to the rear window defogger on vehicles so equipped, which eliminates the need for a separate control and timer module for the rear window defogger system. (Refer to 8 - ELECTRICAL/WINDOW DEFOGGER - DESCRIPTION).
- **Vacuum Fluorescent Display Synchronization** - The EMIC transmits electronic panel lamp dimming level messages which allows all other electronic modules with Vacuum Fluorescent Display (VFD) units on the PCI data bus to coordinate their illumination intensity with that of the EMIC VFD unit.

The EMIC houses six analog gauges and has provisions for up to seventeen indicators (Fig. 3). The EMIC includes the following analog gauges:

- **Coolant Temperature Gauge**
- **Fuel Gauge**
- **Oil Pressure Gauge**
- **Speedometer**
- **Tachometer**
- **Voltage Gauge**

Some of the EMIC indicators are automatically configured when the EMIC is connected to the vehicle electrical system for compatibility with certain optional equipment or equipment required for regula-